CSCI 4150 project checkoff: ECOllateral

Status against the supplied two-page course handout. Presentation: October 20 or 23, 2026; assigned day and Submitty deadline are unconfirmed. The prototype and experiments cover cooling consumption; regional assessment remains planned.

Project Scope & AI Integration

Item	Status	Project response / evidence
Clear goal, problem, and primary question	Documented	Assess development impacts on seasonal area reserves and watershed conditions; cooling is one pressure component
Achievable sub-goals and scope before mid-term	Documented	Working component prototype, four approach investigations, explicit regional scope and remaining area-assessment work
Class AI tools/concepts mapped or exclusions justified	Partial	Explain rules, tree models, word/vector search, prediction ranges, and missing data; wait on neural/LLM work; confirm actual studio names/links

Data Acquisition & Governance

Item	Status	Project response / evidence
Identify public data, permitted scraping, or survey requirements	Documented for current scope	USGS/NOAA public APIs/files; no surveys or private scraping; check future permit sources
Exactly document where data come from and mark public datasets	Documented	`../../data-contracts.md`: exact download URLs, station IDs, dates, and evidence file fingerprints
Confirm instructor permissions for scraping / IRB before surveys	Pending if scope expands	No approval is claimed; Scott coordinates instructor clarification before new scraping or surveys
Clean flow into system components	Implemented / tested	Download -> monthly inputs -> fill gaps from training data -> predict -> check limits -> find permit quotes -> return JSON

System Architecture & Evaluation

Item	Status	Project response / evidence
Visual block diagram of component interactions	Prepared	Regional goal plus current number/permit components; editable diagram in slides; area balance remains planned
Evaluation metrics for individual components	Documented / tested	Units and bad-input checks; error/fit (MAE/RMSE/R2); range coverage; missing-data spread in MGD2; eligible/ineligible locations
Explicit success and failure criteria for outputs	Documented	Enforce limits/warnings; R2 > 0.80 and MAE < 0.05 MGD component goal; >=90% of missing-data changes within tolerance plus band checks; regional/facility criteria unverified
Component test plan	Implemented with limits	October 7 snapshot: 51 passing tests covering forced wrong values, saved-model reload, train-only gap filling, bad neighbors, permit locations, API errors; real-permit relevance testing remains planned

Team Organization & Workflow

Item	Status	Project response / evidence
Assign sub-goals and responsibilities to each member	Assigned	Shaun: code and area integration; Troy: environmental data and grounding; Scott: evaluation, regional evidence, and slides
Shared GitHub repository with instructor/TA access	Partial	Configured remote: `https://github.com/ScottLiu12/ECOllateral`; actual collaborators, instructor/TA access, and pushed snapshot are unverified

Presentation Preparation

Item	Status	Project response / evidence
Show architecture assessment if full system is incomplete	Prepared	Show the working prototype and what real-world tests remain; diagram explains number and permit paths
Draft mid-semester materials and slides	Prepared	Six slides, editable PPTX, PDF export, four-minute script
Explain the problem	Prepared	Slide 1 and dossier problem/user statement
Explain sub-goals, milestones, and teammate mapping	Prepared	Team snapshot and role-owned roadmap
Identify AI concepts/tools used	Prepared	Engineering rules; RF/XGBoost tree models; weighted word matching (TF-IDF); nearby-station/time gap filling; no generative LLM experiment claimed
Justify AI evaluation and feasibility takeaways	Prepared	Four approach investigations with evidence; clear limits of made-up data and about 88% prediction-range coverage
Explain current state and remaining roadmap	Prepared	Component API/tests work; need regional supply/demand, returns, ecological baselines, checked boundaries/documents, measured targets, and independent range tests
Schedule instructor office hours before presentation	Pending	Scott coordinates a course-approved slot

Outstanding course items

- Assigned presentation day and Submitty deadline.
- Instructor/TA repository access and submitted commit.
- Exact class studio references and completed team contributions.
- Office-hours appointment and rehearsal with all three members.